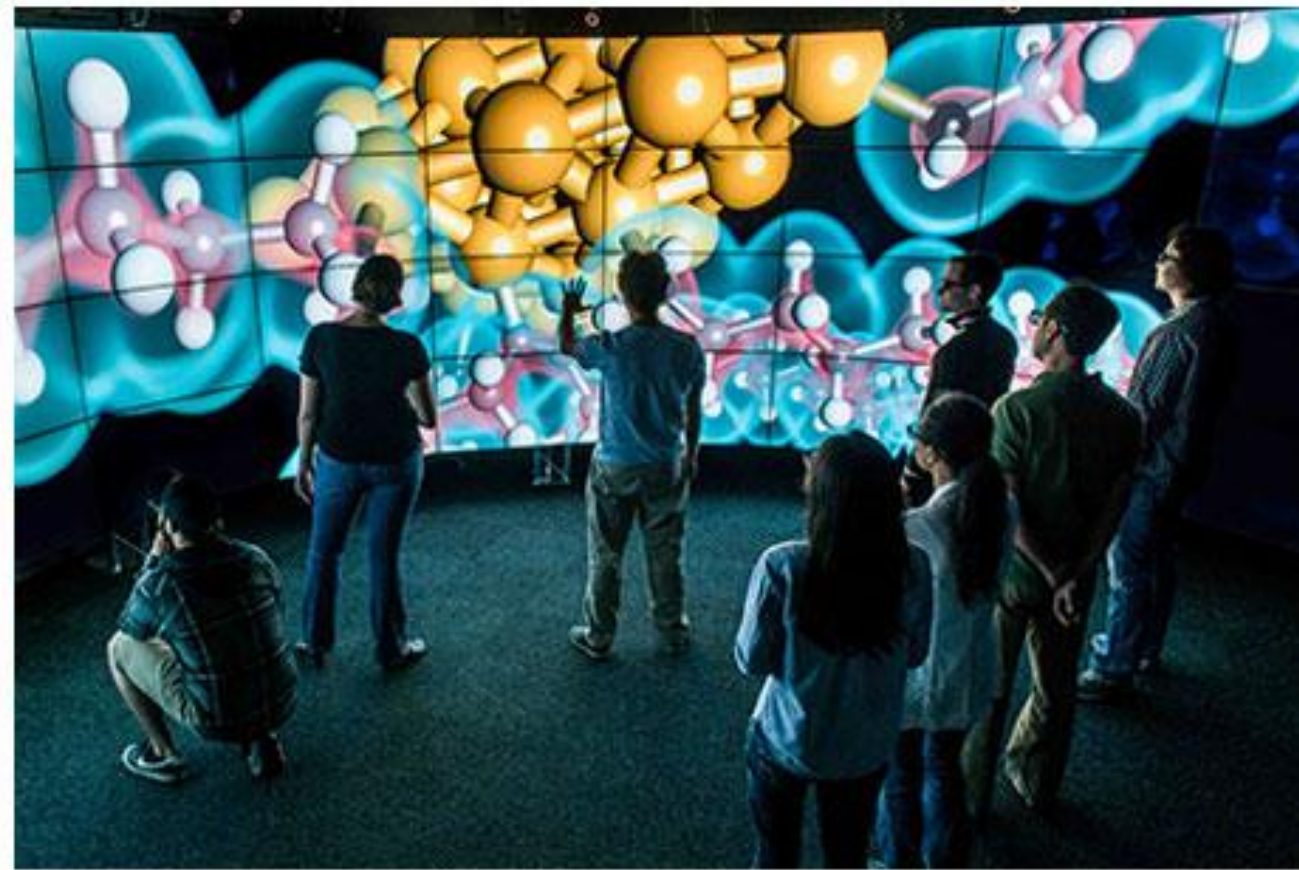




Chemical Engineering



Industrial Advisory Board Meeting
August 14, 2020



Welcome Advisory Board Members!

Agenda

9:00-9:10 Welcome (Alan Zdunek)

9:10-9:45 State of the Department presentation (Vikas Berry)

9:45-10:15 Program Educational Objective review, ABET Fall accreditation overview, self-study report results

10:15-10:45 Curriculum changes (new concentrations, new courses, new 4+1, 3+2, MS Fast track)

10:45-11:00am Wrap-up



State of the Department

Professor Vikas Berry



ABET Assessment: Fall 2020

ABET review will be conducted remotely this Fall.

EAC Team Leader: *Professor Oscar Barton*, Dean of Engineering, Morgan State University

Chemical Engineering Reviewer: *Professor David Tamasko*, Professor and Associate Dean, Ohio State University

Current Status:

(1) Self-Study report (268 pages) completed and sent to ABET (June 2020)

(2) CHE Online System for Fall Review:

- Completing revisions from 2014 online system
- Completing 2015-2020 content additions

(3) Fall Review scheduled for October 2020 (exact date TBD)



Results from ABET Self-Study Report

Format: ABET Criteria form the basis of the assessment and the sections of the Self-Study report:

Criterion 1: Students

Criterion 2: Program Educational Objectives

Criterion 3: Student Outcomes

Criterion 4: Continuous Improvement

Criterion 5: Curriculum

Criterion 6: Faculty

Criterion 7: Facilities

Criterion 8: Institutional Support



Current Program Educational Objectives:

PEO	Category	Description
1	Depth	In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation.
2	Breadth	Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education.
3	Professionalism	In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment.
4	Life-long learning	Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership.

Student Outcome Assessment Method

Method	Description (Abbreviation)	Category	Schedule	Responsibility
1	Grading rubrics in CHE courses (Course Rubrics)	Direct / Formative	Selected courses each year	All faculty
2	Senior Written Examination	Direct / Summative	Fall semester before graduation	DUGS/Undergraduate Committee
3	Engineering Expo judging rubrics (Expo Rubrics)	Direct / Summative	Spring semester	Technical Judges
4	Senior Exit Survey	Indirect / Summative	Spring semester	DUGS/Undergraduate Committee
5	Course Update Surveys (Course Survey)	Feedback	Each semester	All faculty

METHODS 1, 2 and 3:

Performance indicators (PIs) are used to assess students on a major assessment assignment (e.g., exam or portion of exam, term paper, lab report, milestone presentation). PIs are mapped to some or all of the relevant SOs for that course.

Question #	Performance Indicator (PI)	SLO	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
4	6. Description of numerical method and formulation of algorithm	1,2	Does not know how to proceed and/or unable to get started.	Describes most of numerical method but shows some gaps in understanding or is cannot formulate algorithm.	Describes numerical method correctly and successfully formulates algorithm.

SO	Description
1	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3	An ability to communicate effectively with a range of audiences
4	An ability to recognize ethical And professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies

[illegible]

Direct Assessment Results:

The rubric assessment levels are given numerical values of: Satisfactory = 3, Developing = 2, and Unsatisfactory =1. We are looking to achieve a rating score above about 2.30, 65% or better attainment of Satisfactory, and 85% or better attainment for Satisfactory or Developing for each Student Outcome among our students.

Percentage of students attaining a rating of Satisfactory for all the SO's as determined by direct assessment (aggregate of Methods #1, #2, #3).

SO	2020	2019	2018	2017	2016	2015
1	67.2	64.8	69.0	66.8	59.6	64.9
2	64.9	81.2	74.8	74.3	56.0	67.9
3	76.2	82.5	80.8	76.6	70.0	81.3
4	95.2	78.7	90.3	77.4	84.6	78.6
5		91.9	73.8	86.2	88.9	73.4
6		73.9	79.3	66.9	66.7	69.6
7		77.1	72.6	69.6	78.3	87.2

Percentage of students attaining a rating of Satisfactory or Developing for all the SO's as determined by direct assessment (aggregate of Methods #1, #2, and #3).

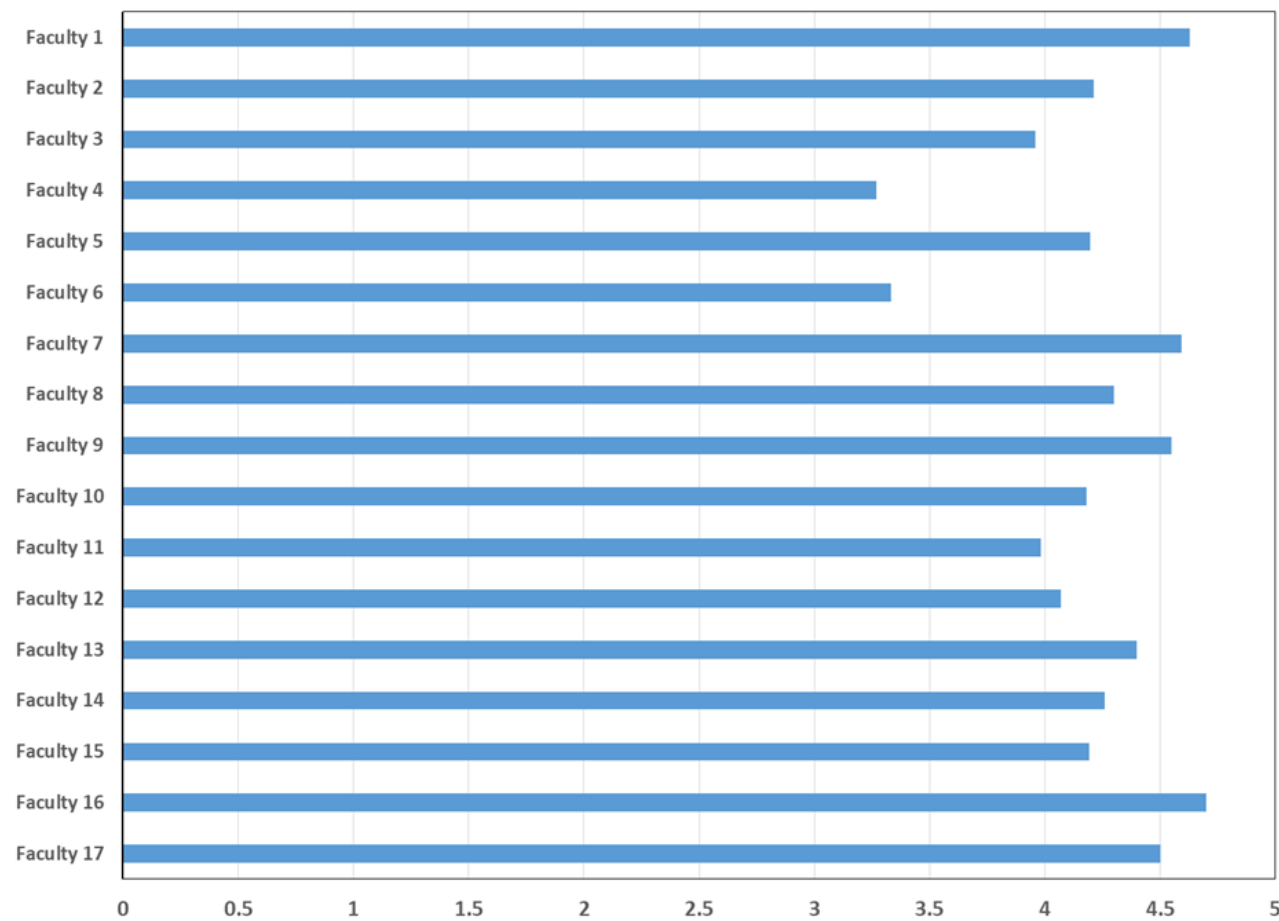
SO	2020	2019	2018	2017	2016	2015
1	86.6	87.2	91.9	91.2	90.7	91.8
2	99.3	97.9	97.1	97.1	84.8	94.0
3	98.9	98.4	97.8	98.8	99.3	99.8
4	98.4	98.5	96.7	97.6	96.4	92.1
5		99.5	96.8	99.1	98.3	97.7
6		99.1	98.3	100.0	98.6	98.7
7		99.2	95.9	99.3	95.5	100.0

Average rating score for all the SO's as determined by direct assessment (aggregate of Methods #1, #2, and #3).

SO	2020	2019	2018	2017	2016	2015
1	2.51	2.44	2.56	2.52	2.33	2.54
2		2.76	2.72	2.72	2.13	2.52
3	2.86	2.76	2.77	2.69	2.72	2.80
4	2.94	2.74	2.84	2.72	2.76	2.71
5		2.79	2.73	2.76	2.73	2.79
6		2.76	2.72	2.69	2.68	2.72
7		2.56	2.55	2.76	2.70	2.80

Method 4 and 5: Course Evaluation and Senior Exit Survey

Average course evaluations for CHE faculty
for the last five years (2015-2020)



Average rating score for all the SO's from Senior Exit Survey

*Note: a different rating system is used than Methods #1- #3:
(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)*

SO	2020	2016	2015
1	1.47	1.50	1.44
2	1.61	1.62	1.60
3	1.40	1.45	1.46
4	1.30	1.26	1.50
5	1.45	1.39	1.60
6	1.54	1.63	1.54
7	1.51	1.57	1.75

Feedback-Based Improvements:

- **Technical elective courses added:** CHE 425 Nanotechnology for Pharmaceutical Applications, CHE 451 Renewable Energy Technologies
- **CHE 381/382 Laboratory changes (student feedback)**
 - New experiments introduced: Gas absorption, continuous distillation, process control, Upgraded equipment, Lab manual revisions
- **CHE 499 Senior Seminar Course revisions (student feedback)**
 - Invited speakers from industry, alumni, career center
 - Written exam instead of oral exam
- **Ethics content added to curriculum (industry mentors/EPC feedback)**
 - CHE 201, 341, 396, 397, 433, 499
- **Safety content strengthened (industry mentors)**
 - SACHE tutorials in CHE 381, 382, 396, 397
 - HAZOP in CHE 396/397, safety assignments in CHE 381, 382
- **ENGR100/Freshmen Engineering Success Program improvement**
 - Technical project breakout sessions in Chemical engineering, GPIIP
- **Extracurricular student learning experiences (student feedback)**
 - ChE Industry Day, ChemE Forum, enhanced student/industry, student/alumni interaction



ChE Online Assessment and Improvement System



Curriculum Changes

Approved BS Degree Curriculum Changes

Five new concentrations (COE approved, UIC SCEP pending):

1. Polymers and Molecular Engineering
2. Energy and Environment
3. Nanotechnology
4. Process Automation
5. Entrepreneurship

Six new courses (Approved February 2020):

CHE 230: Molecular Systems in Chemical Engineering

CHE 330: Polymer Science

CHE 453: Fundamentals of Electrochemistry

CHE 454: Molecular and Macromolecular Engineering

CHE 455: Nanoscale Systems in Chemical Engineering

CHE 457: Colloidal and Interfacial Phenomena

Selective course option : CHE 330 or CHEM 346

400-level courses converted to 500-level (graduate level): CHE 410 (Transport Phenomena), CHE 431 (Numerical Methods in Chemical Engineering), CHE 445 (Mathematical Methods in Chemical Engineering)

Approved BS/MS Degrees

4+1 BS/MS program

Earned separately, the degrees require six years and a total of 164 hours, 128 hours for the B.S. and 36 for the M.S. The joint degree can be completed in five years, and students earn a total of 158 hours. The requirements for completion of the joint degree are identical to the requirements for the completion of the two separate degrees; however, eight hours of shared coursework applies to both degrees.

3+2 BS/MS program

Our 3 2 China program allows students from specific universities in China to spend the fourth and fifth years of their university education at the University of Illinois Chicago, culminating in both a bachelor's degree from their home university and an MS in Chemical Engineering from UIC.

Fast Track CHE MS for Non-ChE undergraduates:

Students who choose the one-year fast-track format have the option to pursue the project option or to complete the degree entirely with coursework.

One-year fast-track students who pursue the project option spend a fall, spring, and summer semester at UIC, meaning that they begin classes in August and graduate the following August. One-year students who choose the coursework-only option complete their degrees in one fall and one spring semester, starting classes in August and graduating in May.



Curriculum Changes in Progress

- CHE 150 Climate Engineering for Global Warming (General Education Course)
- CHE 101 Introduction to ChE
- Revisions due to Chemistry department course credit changes (-2 hour deficit):
 - CHEM 232: 4 hours → 3 hours
 - CHEM 234: 4 hours → 3 hours

Two proposals

Change	Credit hours gained	Credit hours lost
Add CHE 101	+3	
Add CHE 230	+3	
Change CHE 396 hours 4 → 3 hours		-1
CHEM 232/234 credit hour change		-2
Remove ECE 210 (Electrical Circuit Analysis)		-3

Change	Credit hours gained	Credit hours lost
Add CHE 101	+3	
Add CHE 230	+3	
Change CHE 396 hours 4 → 3 hours		-1
CHEM 232 credit hour change		-1
Remove CHEM 234		-4



Wrap-Up

- Summary and Action items
- Next meeting: Summer 2021



Chemical Engineering Industrial Advisory Board

Thank you for your time